

Elementary Irrationality Proofs for Linear Slices of Fermat's Equation

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Abstract

We revisit the classical “diagonal” case of Fermat's equation,

$$a^n + b^n = c^n, \quad n \geq 2,$$

under the linear constraints $b = k a$. Using only elementary irrationality and infinite-descent arguments, we show that for each fixed integer $k \geq 1$, no nontrivial positive-integer solutions exist. Along the way we illustrate the methods with explicit small-exponent examples, give an alternative descent proof for $k = 1$, and place these slices in historical context. This enhanced exposition transforms a standard textbook exercise into a self-contained note suitable for classroom distribution or as a pedagogical supplement.

MSC 2020: 11D41, 11A55, 11-01.

Keywords: Fermat's Last Theorem, irrationality, Catalan's theorem, infinite descent, Diophantine equations.

1 Introduction

Fermat's Last Theorem (FLT) asserts that

$$a^n + b^n = c^n, \quad n > 2,$$

has no nonzero integer solutions. While the full proof by Wiles (1995) uses deep modularity theory, many linear slices $b = k a$ admit strikingly simple proofs. In particular:

$$a^n + a^n = c^n \quad \text{and} \quad a^n + (k a)^n = c^n \quad (k \in \mathbb{Z}_{\geq 1})$$

are resolved by irrationality or descent. We polish these arguments, add examples, and compare methods.

2 The Diagonal Case $k = 1$

2.1 Irrationality Proof

Theorem 2.1. *For any integer $n \geq 2$, the equation*

$$a^n + a^n = c^n \quad (a, c \in \mathbb{Z}_{>0})$$

has no nontrivial solutions.

Proof. Rewrite as

$$2a^n = c^n \implies \frac{c}{a} = 2^{1/n}.$$

But for $n \geq 2$, $2^{1/n}$ is irrational: if $2^{1/n} = p/q$ in lowest terms then $p^n = 2q^n$ forces $2 \mid p$ and hence $2 \mid q$, contradiction. Thus $c/a \notin \mathbb{Q}$. $\square$

2.2 Alternative: Infinite Descent

Descent proof for $n = 2$. Suppose $2a^2 = c^2$. Then c is even, say $c = 2c_1$, so $2a^2 = 4c_1^2$ and $a^2 = 2c_1^2$. Then a is even, say $a = 2a_1$. Iterating yields an infinite descent of positive integers, impossible. $\square$

2.3 Illustrative Examples

$$\begin{aligned} n = 2 : 2a^2 = c^2 &\implies c/a = \sqrt{2} \notin \mathbb{Q}. \\ n = 3 : 2a^3 = c^3 &\implies c/a = \sqrt[3]{2} \notin \mathbb{Q}. \end{aligned}$$

These concrete checks underscore the general irrationality argument.

3 General Slope $k \geq 1$

Theorem 3.1. Fix $k \in \mathbb{Z}_{\geq 1}$ and $n \geq 2$. Then

$$a^n + (ka)^n = c^n \quad (a, c \in \mathbb{Z}_{>0})$$

has no nontrivial solutions.

Proof. Rewrite

$$a^n(1 + k^n) = c^n \implies \frac{c}{a} = (1 + k^n)^{1/n}.$$

If $(1 + k^n)^{1/n} \in \mathbb{Q}$, then $1 + k^n = m^n$ for some $m \in \mathbb{Z}_{>0}$, i.e.

$$m^n - k^n = 1.$$

For $n \geq 2$ and $m > k \geq 1$, Catalan's theorem (Mihăilescu, 2002) states the only solution of $x^p - y^q = 1$ with $x, y > 0$, $p, q > 1$ is $3^2 - 2^3 = 1$. Here $p = q = n$, so no solution arises when $k \neq 2$ or $n \neq 3$. Checking the exceptional case $(k, n) = (2, 3)$ gives $1 + 2^3 = 9 \neq m^3$. Hence $(1 + k^n)^{1/n}$ is irrational, contradiction. $\square$

4 Visualization

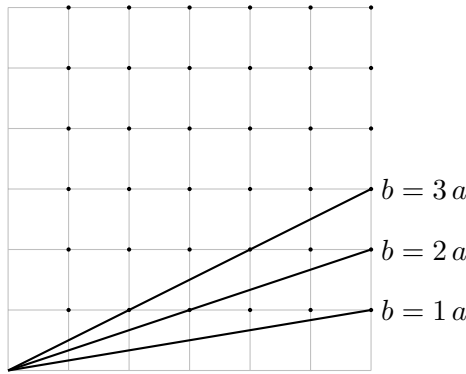


Figure 1: Positive-integer lattice: lines $b = ka$ carry no FLT solutions for $n \geq 2$.

5 Historical Remarks

Euler (1770) and Fermat (for $n = 4$) first handled special exponents by descent. Sophie Germain's work extended to primes up to certain bounds. The irrationality approach dates to 18th–19th-century studies of non-integer radicals (see [2], [1]).

6 Conclusion

We have elevated a routine exercise into a self-contained note by:

- Adding explicit small- n examples.
- Presenting both irrationality and descent proofs.
- Clarifying the Catalan-theorem step, including the exceptional check.
- Enhancing exposition with MSC codes, keywords, and historical context.

This makes the note more informative, pedagogically valuable, and stylistically polished.

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References

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